

Manufacturing Junction gateKeeper AI Vision SOP

SOP-FAV-001 | Offline field installation, Frame Gate setup, labeling and CPU fine-tuning | Revision 1.1.1

1. Purpose and safety boundary

Purpose

Read the FPCB code before AAM input. Results are **OK**, **Abnormal**, **Error** (registered problem code), or **System Error**.

Default recipe

Expected normal: **HJ04**. Registered problem code: **HJ05**. Only O/O is corrected automatically.

Field trial safety: keep PLC in Simulation or Dry-run. Do not connect a stop, reject, or picker output until the site safety review and independent real-lot validation are approved.

2. Install and open the offline application

1. Copy the release ZIP to a local SSD, for example `D:\gateKeeper`. Do not run from USB or a network share.
2. Extract all files. Keep the extracted folder together; do not move `_internal`, `models`, `config`, or `docs`.
3. Run **Manufacturing Junction gateKeeper AI Vision.exe**. Obtain local IT approval if Windows shows a security warning.
4. The first hot-folder start loads the CPU detector and OCR model in the background. The window must remain responsive.

Pass condition: the app shows HJ04/HJ05. After starting the folder it shows **LIVE / CPU / HOT-FOLDER / FRAME GATE**, not System Error.

3. Select the camera folder and start Frame Gate

1. Create or select one local camera output folder, for example `D:\gateKeeper\camera-watch`. Configure the camera to save unique final JPG/PNG files there.
2. In the dashboard, press **Choose folder**, select that same folder, then press **Start hot-folder**.
3. With the view empty, press **Capture Empty Background**. Open **Frame Gate Settings**; use 150 ms camera interval, two stable frames, two empty frames to rearm, and 2000 ms refractory period as the initial field value.
4. Confirm the blue LIVE text. A selected stationary frame alone proceeds to YOLO/OCR; empty, moving, duplicate, and refractory frames are not production decisions.

Manufacturing Junction gateKeeper AI Vision - AAM FPCB Inspection

Manufacturing Junction gateKeeper AI Vision | AAM FPCB Code Inspection

Inspection controls

Expected code

HJ04

Problem codes

HJ05

Camera hot-folder

C:\gateKeeper\watch

Choose folder

Manage code recipe

Simulation OCR

HJ04

Detection candidate confidence

0.01

Final ROI confidence

0.01

OCR confidence

0.98

OCR ROI relative X

0.00

OCR ROI relative Y

0.00

OCR ROI relative width

1.00

Latest result

Waiting for image

No inspection has been processed

Verified UI reference: Camera hot-folder selection is on the left; Frame Gate and result panel are visible without hiding controls. Scroll the left panel to access labeling and training controls on smaller displays.

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Continue: recipe management, labeling, CPU fine-tuning and safe field trial

4. Verify the result rule before camera trial

Known test input	Required result	Operator action
Clear HJ04	Green OK; no alarm	Confirm archive and overlay were saved.
Clear HJ05	Red Error; repeat alarm; simulated RejectRequest	Follow quality procedure; verify simulation log only.
Blurred, missing ROI, unreadable, or other code	Yellow Abnormal; never green	Inspect image and add feedback if appropriate.
Model/folder/storage failure	System Error	Stop monitoring; never treat it as OK.

5. Manage codes, label real images, and export the dataset

1. Press **Manage code recipe**. Add, edit, or delete only approved four-character uppercase codes. Keep normal and problem lists separate; save the active expected normal code.
2. Open **Image labeling workbench** and press **Open local panel image**. Draw `fpcb_surface` around the FPCB, then draw `code_roi` tightly around the visible code.
3. Select the OCR code, enter panel/lot/recipe group, and press **Save reviewed label**. Use the same group for frames from the same physical panel or lot.
4. Press **Export grouped YOLO dataset**. Fix every validation error. Never place frames from one panel/lot in both training and test splits.



6. CPU fine-tuning and approval

1. In **Training progress**, select the exported dataset and the approved checkpoint. Initial CPU values: 640 image size, batch 1–2, workers 0, 100 epochs, early-stop patience 20.
2. Start CPU training. Check the validation graph, exact-code accuracy, problem recall, confusion matrix, and saved best checkpoint.
3. Test a candidate only on a held-out real lot that was not used in labeling or training. Do not auto-promote a model from operator feedback.

Promotion gate: approve only after independent field holdout results meet the site target, no problem sample becomes normal, ROI overlays are correct, and the PLC fail-safe review is signed.

7. End-of-shift and recovery

Check

- HJ04 green; HJ05 red; blurred/unknown never green.
- Frame Gate suppressions do not affect decision counters.
- Alarm/Mute, archive, overlay, duplicate guard, and latest popup replacement work.

Recover

- On System Error: stop monitoring; check disk, folder permission, model files, and `logs/startup-errors.log`.
- Use `docs/OPERATOR_GUIDE.md`, `docs/TRAINING.md`, and `docs/PLC_INTEGRATION.md` for detail.

Stop and escalate: unexpected green result, wrong ROI, missing archive, System Error, or any request to enable physical PLC outputs without formal safety approval.